

OTT Platform Customer Churn Analysis

OTT Platform Customer Churn Analysis — Project Insights

A SQL + Python driven churn analytics project uncovering why subscribers leave an OTT streaming platform — and where the revenue risk hides.

Project Overview

Subscriber retention is the single biggest lever for profitability in the OTT/streaming business — acquiring a new subscriber costs far more than keeping an existing one. This project simulates a real-world OTT backend (customer profiles, subscription/billing records, and support tickets stored in a relational SQLite database) and builds a complete churn analytics pipeline: from raw, messy multi-table data → cleaned & merged master dataset → engineered churn features → exploratory analysis → actionable business insights.

Goal: Identify *who* is churning, *why*, and *how much revenue* is at stake — the way a Data/Business Analyst would for a streaming subscription business.

Data Architecture

The raw data lived across **3 relational tables** inside a SQLite database (`customer_churn.db`), mirroring how a real OTT platform’s backend is typically structured:

Table	Contents
<code>db_customer</code>	Subscriber demographics — name, country, state, gender, DOB, interests
<code>db_subscription</code>	Plan type, contract type, billing, renewal/cancellation dates, CLTV, churn score
<code>db_support</code>	Complaint history, escalations, CSAT scores

These were queried via SQL (`sqlite3` + `pandas.read_sql`), inspected with `PRAGMA table_info` , and merged into a single **852-row, 21-feature master dataset** for analysis.

Data Cleaning & Feature Engineering

Real subscriber data is never clean — this project handled the kind of inconsistencies analysts see daily:

- Standardized categorical noise** — merged inconsistent gender labels (`Men / Women` → `Male / Female`)
- Smart missing-value imputation** — filled missing `country` values using a `state → country` mapping instead of dropping rows
- Date normalization** — converted subscription, renewal, cancellation, DOB, and complaint dates to proper `datetime` types
- Churn labeling** — engineered a binary `churn_flag` from `cancellation_date` (churned vs. active)
- Tenure calculation** — computed `tenure_days` for every subscriber (time active, or time-to-cancellation)
- Risk segmentation** — bucketed subscribers into `low / medium / high` churn risk tiers based on `churn_score`
- Support signal integration** — aggregated complaint counts and escalation flags per customer and joined them into the master table
- Categorical encoding** — ordinal-encoded plan tier, contract type, and churn risk for correlation analysis

Key Business Metrics

Metric	Value
Overall Churn Rate	14.08%
Retention Rate	85.92%

ARPU (Avg. Revenue per User)	\$15.34/month
Average Subscriber Tenure	~1,555 days
Revenue at Risk (from churned users)	\$1,873.12K
Support Escalation Rate	11.97%
Avg. Complaints per Customer	0.81

Churn Rate by Plan Type

Plan	Churn Rate
Premium	15.65%
Standard	13.65%
Basic	13.33%

→ Counter-intuitively, **Premium subscribers churn the most** — a red flag that high-tier customers aren’t seeing enough value for what they pay.

Churn Rate by Acquisition Channel

Channel	Churn Rate
Referral	15.35%
Paid	14.97%
Organic	11.85%

→ **Organic sign-ups are the stickiest**, while paid and referral acquisition bring in less loyal subscribers — a key input for marketing spend allocation.

Churn Rate by Geography

State-level churn ranged from as low as **2.94% (Odisha)** to as high as **27.27% (Meghalaya)** — a nearly **9x variance** — revealing that regional service quality, content localization, or competitor presence likely play a major role in retention.

🔍 Exploratory & Visual Analysis

Using **Matplotlib** and **Seaborn**, the project visualizes churn from multiple angles:

- 📈 **Monthly churn trend line chart** — tracking cancellations over time to spot seasonality/spikes
- 📊 **Churn rate by plan type & by state** — bar charts highlighting the highest-risk segments
- 🔥 **Correlation heatmap** — quantifying relationships between plan type, contract type, churn score, and churn risk
- 🔗 **Pairplot & category-faceted plots** — examining how monthly charges, plan type, and gender interact with churn risk tiers

💡 Business Recommendations

- Investigate the Premium tier experience** — with the highest churn despite the highest price point, this is the biggest revenue leak.
- Prioritize retention campaigns in high-churn states** (Meghalaya, Nagaland, Assam) where churn exceeds 20%.
- Proactively target “high churn-risk” subscribers** (churn score ≥ 70) with retention offers before they cancel.
- Rethink referral/paid acquisition incentives** — organic subscribers show meaningfully better retention.
- Tighten the support-to-retention loop** — with ~12% of subscribers escalating issues, faster resolution could directly reduce cancellations.

🔧 Tech Stack

Python · Pandas · NumPy · SQLite3 · Matplotlib · Seaborn · SQL

Skills Demonstrated

SQL data extraction from a multi-table relational database · data cleaning & imputation · feature engineering · churn labeling & risk scoring · cohort/segment analysis · correlation analysis · data visualization · business-focused storytelling

This project reflects the end-to-end workflow of a Data Analyst supporting subscription/retention strategy for a streaming (OTT) business — from raw backend data to decision-ready insights.